

README - Python IMU Logger

This guide explains how to set up Python and use the IMU Logger tool to receive and visualize data from the "MainBoard_IMU_Logger" example project.

Install Python 3.12 or newer

1. Go to the official website: <https://www.python.org/downloads/>
2. Click "**Download Python 3.12.x**" (or the latest available version).
3. During the installation, **make sure to check**: "Add Python to PATH". This step is very important as it allows you to run Python from the terminal.
4. Complete the installation by following the default steps.

Install Visual Studio Code (VS Code)

1. Go to: <https://code.visualstudio.com/download>
2. Download and install **VS Code** for your operating system (Windows/macOS/Linux).
3. When installation is complete, open VS Code.
4. Inside VS Code, install the following extensions (recommended):
 - **Python** (by Microsoft)

Open the project in VS Code

1. Open VS Code.
2. Click on **File → Open Folder...**
3. Select the project folder "IMU_Logger_Python"
4. Inside the Explorer panel, you should see:

```
IMU_Logger_Python/  
├─ download_script.py
```

- └ requirements.txt
- └ README - Python IMU Logger

Create and activate a virtual environment (venv)

1. Open the VS Code terminal.
2. Create a virtual environment inside the project folder:

```
python -m venv venv
```

1. Activate the virtual environment:

- **Windows PowerShell:**

```
.\venv\Scripts\Activate.ps1
```

- **Windows CMD:**

```
.\venv\Scripts\activate.bat
```

- **macOS/Linux:**

```
source venv/bin/activate
```

You should see (venv) at the beginning of your terminal prompt.

All Python packages will now be installed inside this isolated environment.

Install the required libraries

1. Open the VS Code terminal
2. In the terminal, type the following command:

```
pip install -r requirements.txt
```

This will automatically install all the required Python packages:

- `numpy`
- `matplotlib`
- `pandas`
- `pyserial`

If you see an error, try upgrading pip first:

```
python -m pip install --upgrade pip
```

Run the program

1. In the VS Code terminal, run the script:

```
python download_script.py
```

A small window will open:

- Select the correct **COM port** (e.g., `COM3`, `COM4`, etc.)
- Choose the **destination folder** to save the data
- Click  **Confirm and Start**
- You should see the code messages in the Python terminal